Faculty of Engineering
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CMPS451: Big Data

Project Proposal: Smart Music Recommendation System Based on User's Choices – Part 2

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A) Datasets

Note: the sizes mentioned are the sizes of the files that should be used in this project, not the full sizes of the datasets

https://www.kaggle.com/datasets/tomigelo/spotify-audio-features - size: 38.47 MB

https://www.kaggle.com/datasets/mrmorj/dataset-of-songs-in-spotify - size: 13.6 MB

https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset - size: 20.12 MB

https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks-db - size: 33.71 MB

https://www.kaggle.com/datasets/tonygordonjr/spotify-dataset-2023

http://millionsongdataset.com/ - size: The actual size is 280 GB, we'll try to use as much as possible from it with minimum amount of use of 1.8 GB

B) Planned approach

1. **Data Preprocessing (using Apache Spark):**
   Load and clean the dataset, select relevant audio features, and normalize numerical values.
2. **Exploration & Visualization:**
   Analyse feature distributions and relationships to better understand the data.
3. **Similarity-Based Recommendation:**
   for song representation: each song is represented as a feature vector and compute similarity using **cosine similarity**, A similarity matrix will be precomputed for efficiency.
   For the advanced approach, ALS will be used to model user-song interactions.
   Playlist data will be used to build a user-item interaction matrix, where each playlist is treated as a user and songs represent items. If a song appears in a playlist, it is considered an interaction.
   ALS will then learn patterns between playlists and songs by discovering hidden relationships (latent factors). Based on these patterns, the system can recommend songs that are likely to fit the same playlist or user

4. **Playlist Generation:**
   Given an input user, the system retrieves the top N most similar songs to his taste and generates a playlist.
5. **Evaluation:**
   Evaluate the quality of recommendations and compare results with alternative methods (e.g., KNN) if needed.